

ShadowEvent AI

Transforming Recurring Traffic Failures into Predictable Events

Problem Statement

Traffic authorities face two major categories of disruptions:

Planned Events

- Political rallies
- Festivals
- Sports matches
- Road construction activities
- Public gatherings

Unplanned Events

- Recurring congestion hotspots
- Frequent accident zones
- Peak-hour traffic breakdowns
- Localized traffic incidents

Current traffic management systems primarily focus on reacting to incidents after they occur. Even when historical data exists, recurring traffic failures are rarely treated as predictable patterns.

As a result:

- Resource deployment is largely experience-driven.
- Recurring hotspots continue to create congestion.
- Historical incident knowledge remains underutilized.
- No structured mechanism exists to proactively prepare for recurring traffic failures.

Core Insight

Not all "unplanned" events are truly unplanned.

Many traffic incidents occur repeatedly at the same locations, during the same time windows, and under similar conditions.

Examples:

- Silk Board every weekday evening
- Mekhri Circle every Friday peak hour
- Airport Road Monday morning rush

These recurring disruptions behave like scheduled events despite not being officially announced.

Proposed Solution

ShadowEvent AI

ShadowEvent AI identifies recurring traffic patterns from historical incident records and converts them into:

Shadow Events

A Shadow Event is a recurring traffic disruption that exhibits predictable temporal and spatial patterns.

Instead of waiting for congestion to emerge, authorities can proactively prepare for these hidden recurring events.

System Overview

Historical Traffic Incident Data ↓ Pattern Mining Engine ↓ Shadow Event Discovery ↓ Risk Calendar Generation ↓ Impact Zone Analysis ↓ Resource Planning Dashboard ↓ Knowledge Repository

Module 1: Shadow Event Discovery Engine

Objective

Identify recurring congestion and incident patterns hidden within historical traffic data.

Methodology

Analyze:

- Location
- Day of week
- Time of day

- Incident category
- Historical frequency

Compute recurrence statistics for:

Location × Day × Hour

Example Output

Location: Silk Board

Time Window: Monday-Friday 6 PM - 9 PM

Historical Frequency: 73% occurrence rate

Generated Shadow Event:

"Silk Board Evening Congestion Event"

Value

Transforms recurring failures into predictable operational events.

Module 2: Event Pattern Mining

Objective

Discover recurring incident behaviors and relationships within historical data.

Examples

Recurring Hotspots

- Locations with consistently high incident rates

Recurring Time Windows

- Peak periods for specific incident types

Incident Associations

Example:

Road Closure → Frequently followed by Congestion Event

Output

A ranked list of:

- High-risk locations
 - High-risk time periods
 - Frequently occurring incident patterns
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Module 3: Weekly Risk Calendar

Objective

Convert historical patterns into actionable operational intelligence.

Output Example

Monday

Airport Road Risk Level: High

Tuesday

Location B Risk Level: Medium

Thursday

Silk Board Risk Level: High

Friday

Mekhri Circle Risk Level: High

Benefits

Traffic authorities receive a proactive forecast of likely incident hotspots before congestion develops.

Module 4: Event Similarity Search

Objective

Provide context for newly observed incidents using historical analogs.

Methodology

Compare incidents using available dataset attributes such as:

- Location
- Day of week
- Time window
- Incident category

Nearest-neighbor retrieval identifies similar historical events.

Example

Current Event:

Congestion Silk Board Thursday 7 PM

Similar Historical Events Found:

15 matching historical records

Observed Outcomes:

- High congestion probability
 - Increased incident frequency
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Benefits

Provides evidence-based decision support rather than relying solely on intuition.

Module 5: Impact Zone Visualization

Objective

Visualize areas potentially affected by a planned event or identified Shadow Event.

Data Sources

- Historical incident locations
 - Road network geometry
 - Mapping services (Google Maps/OpenStreetMap)
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Features

Interactive Map

Displays:

- Event location
 - Historical hotspots
 - Risk zones
 - Monitoring areas
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Example

Event: Political Rally

Visualization:

Primary Risk Zone: 1 km radius

Secondary Risk Zone: 3 km radius

Nearby Historical Hotspots: Junction A Junction B Junction C

Important Note

This module does not perform traffic simulation.

It provides data-driven geographic context and impact visualization.

Module 6: Traffic Knowledge Repository

Objective

Preserve and organize historical operational knowledge.

Stored Information

- Incident location
 - Timestamp
 - Incident category
 - Historical frequency
 - Associated patterns
 - Similar past incidents
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Benefits

Creates a searchable institutional memory for future planning and analysis.

Innovation

Innovation 1: Shadow Event Modeling

Traditional systems focus only on officially announced events.

ShadowEvent AI introduces the concept of:

Recurring Congestion Patterns

Predictable Events

This allows proactive planning for disruptions that were previously treated as random.

Innovation 2: Unified Treatment of Planned and Unplanned Events

Most systems separate:

Planned Events

and

Unplanned Incidents

ShadowEvent AI places both within a common operational framework.

Innovation 3: Historical Intelligence Driven Planning

Instead of relying solely on prediction models, the system leverages recurring historical patterns to support operational decision-making.

Technical Stack

Frontend

- React
- Leaflet / Google Maps API

Backend

- Python
- FastAPI

Data Processing

- Pandas
- NumPy

Analytics

- Scikit-Learn
- Clustering
- Nearest Neighbor Search

Visualization

- Plotly
- Folium

- Leaflet

Database

- PostgreSQL / SQLite
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Expected Deliverables (Hackathon MVP)

Deliverable 1

Shadow Event Discovery Dashboard

Displays:

- Recurring hotspots
 - Recurring time windows
 - Recurrence probabilities
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Deliverable 2

Weekly Risk Calendar

Displays:

- High-risk locations
 - Predicted risk periods
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Deliverable 3

Event Similarity Explorer

Displays:

- Similar historical incidents
 - Historical outcomes
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Deliverable 4

Interactive Impact Zone Map

Displays:

- Event locations
 - Historical hotspots
 - Risk zones
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Deliverable 5

Traffic Knowledge Repository

Displays:

- Searchable historical event intelligence
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Future Scope

The following capabilities are intentionally excluded from the MVP because they require additional datasets and validation:

- Traffic Digital Twin Simulation
- Reinforcement Learning Resource Optimization
- Congestion Propagation Modeling
- Dynamic Signal Control
- Traffic Flow Simulation

These can be integrated in future versions when richer traffic and road-network datasets become available.

Elevator Pitch

ShadowEvent AI transforms recurring traffic failures into predictable events. By mining historical incident data, the system discovers hidden congestion patterns, generates proactive risk calendars, visualizes impact zones, and helps authorities prepare for disruptions before they occur. Instead of reacting to traffic incidents, cities can anticipate and manage them using data-driven operational intelligence.